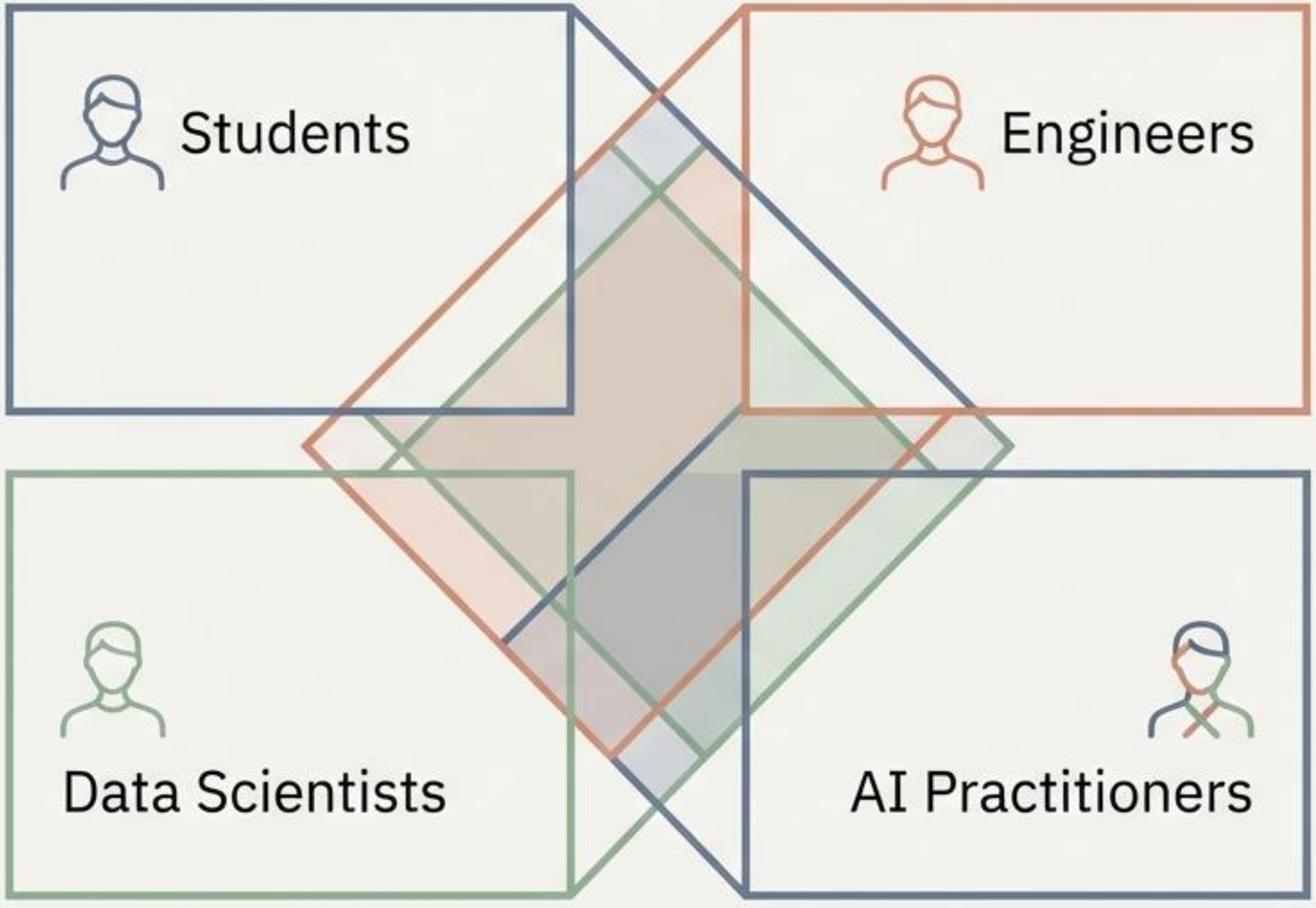
AI / Data in Healthcare Hackathon

Building AI systems that
Understand and reason clinical data

University of Toronto
March 2026

An Experimental Space for AI Practitioners



BRAVE

About Brave Career

Organizing technical events, hackathons, and workshops to help engineers experiment with rapidly evolving tech and build real-world experience through collaboration.

Developing Skills
in the age of AI
Future City Hackathon with FCI & ElevenLabs
Guest Talk By Vyom Malhotra

9th Annual
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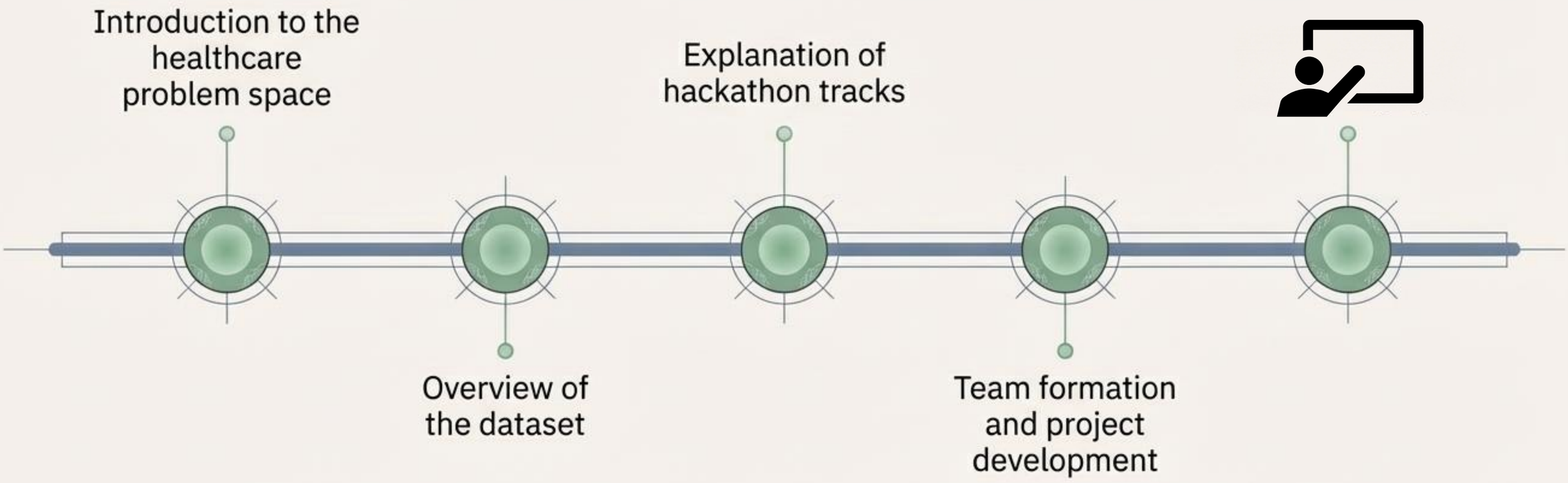
The Blueprint for Today's Event



Priority: Most of the time today is dedicated to building and experimenting.

Important Dates

- Mar. 11th Final Submission
- Mar. 13th Final Presentation & Networking

Judging & Networking Guests

Jason
Associate Director @ AstraZeneca

Suting
Senior Biostatistician @ SickKids

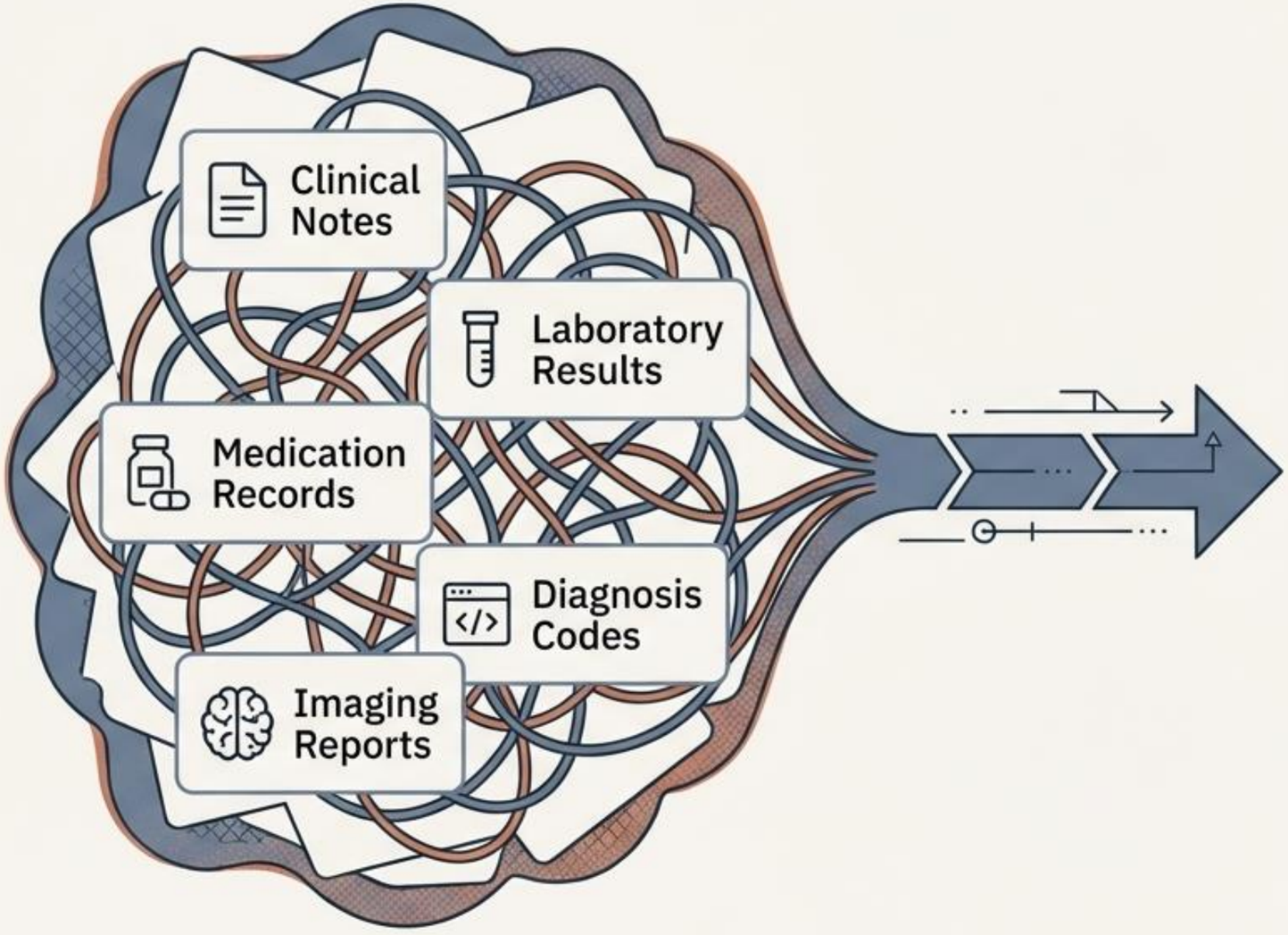
Daisy
Senior Biostatistician @ Tigermed

FINN
Data Analyst @ Siemens Healthineers

Haihui
Research Analyst @ CAMH

Sponsors!

The Overwhelming Scale of Clinical Information



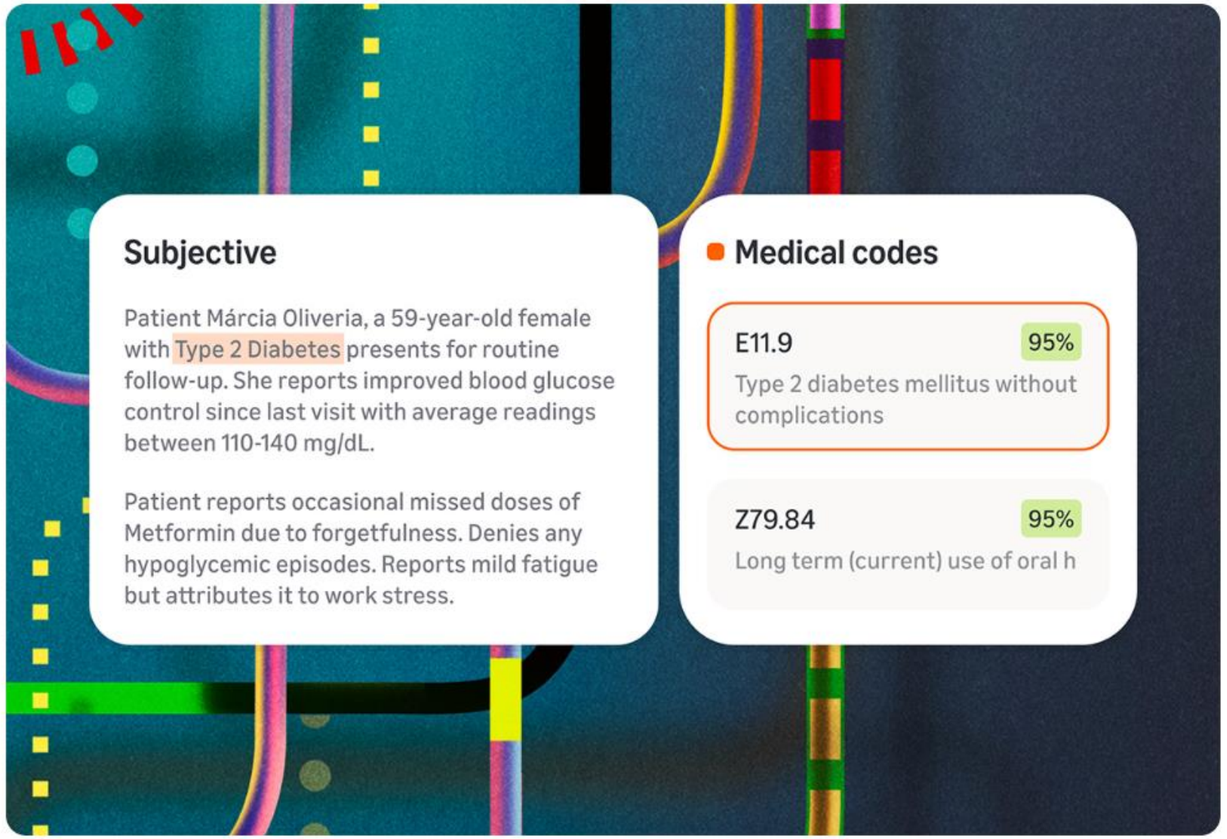
- Clinical Documentation Assistants
- Medical Coding Automation
- Clinical Summarization Systems
- Decision-Support Tools
- AI Copilots for Clinicians

aws

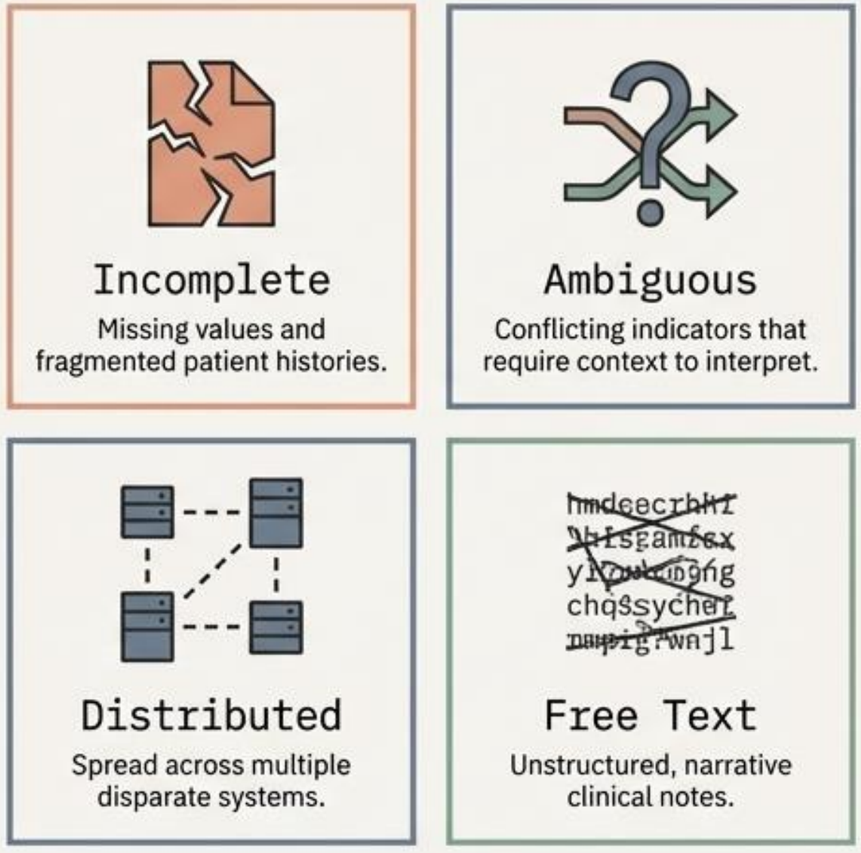
Discover AWS Products Solutions Pricing Resources

Amazon Connect Health

Overview Pricing Partners FAQs

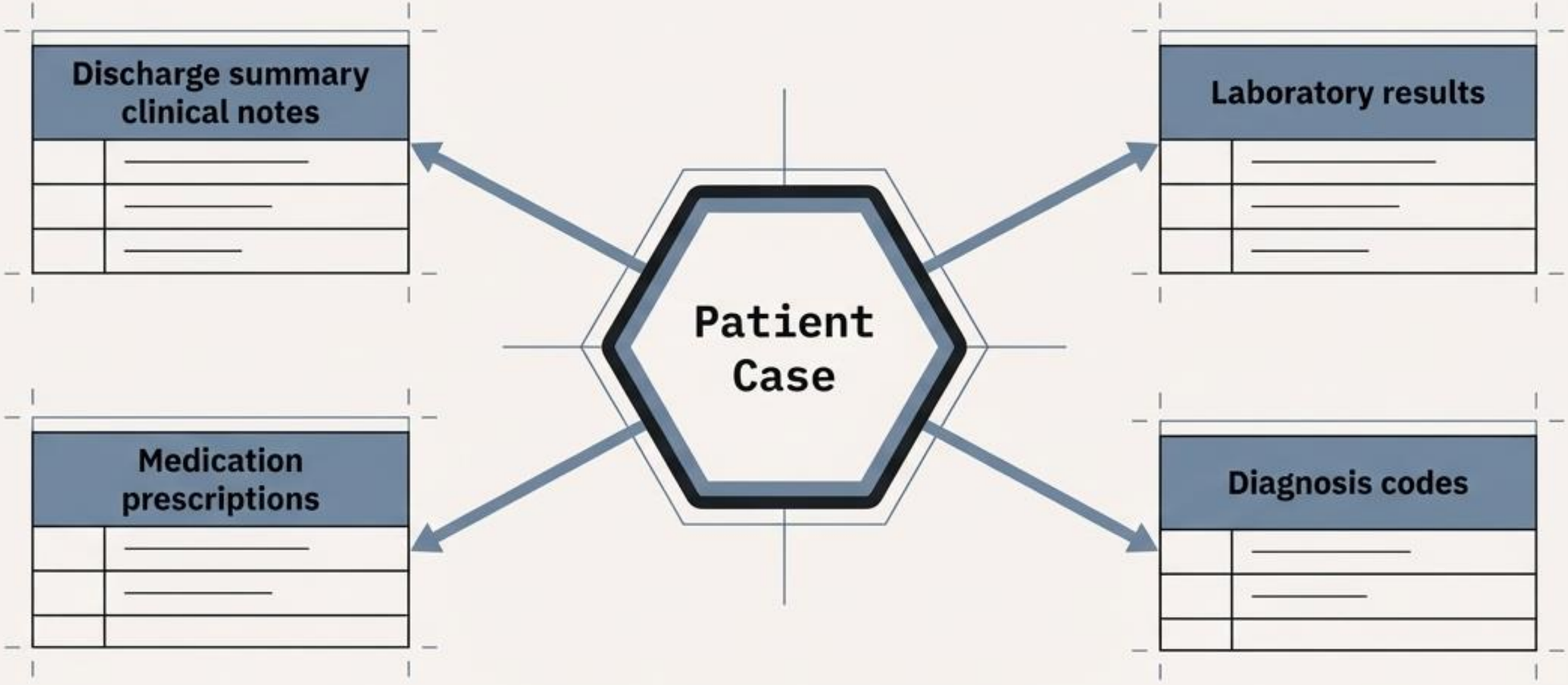


The Reality of Real-World Healthcare Data



Building reliable systems is challenging. The AI must be designed to successfully reason across multiple, imperfect sources of information to produce useful outputs.

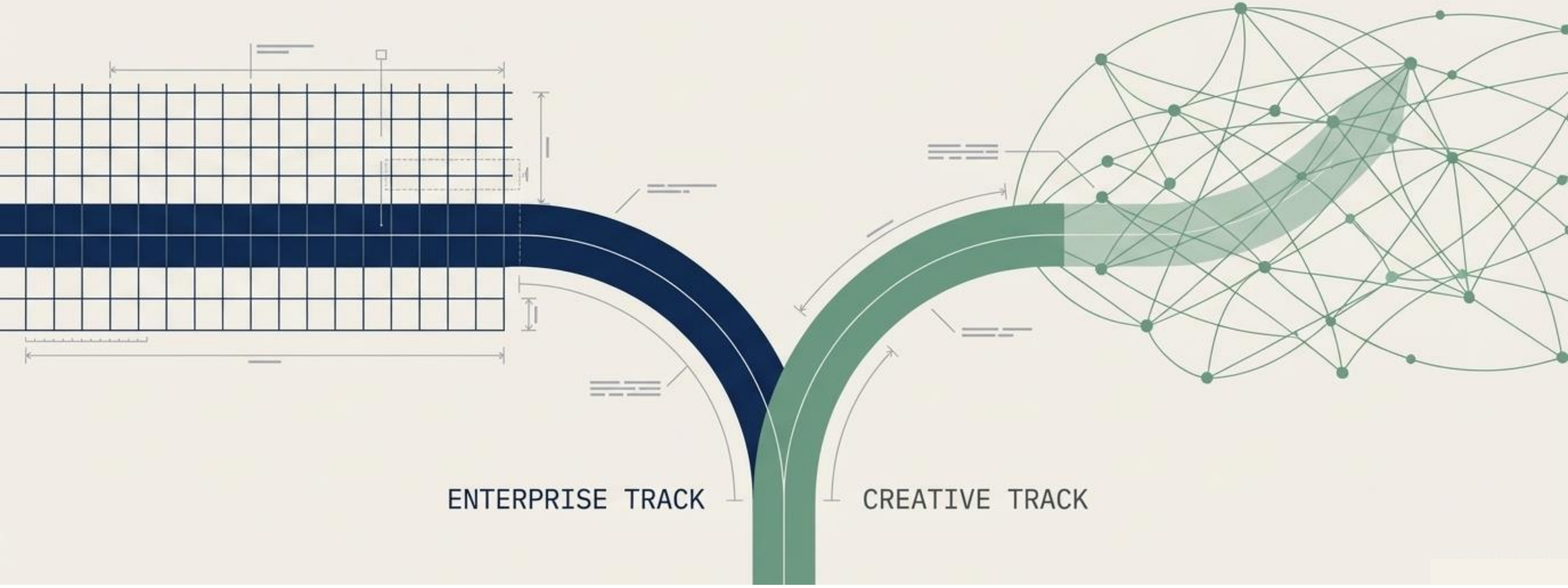
Your Sandbox: 2,000 Relational Hospital Admission Cases



Actionable Insight: The relational structure allows multiple tables to be joined to analyze and reason over complex patient cases.

Choose your trajectory for impact

Participants must **select one of two distinct tracks,**
each with its own specific focus and technical demands.

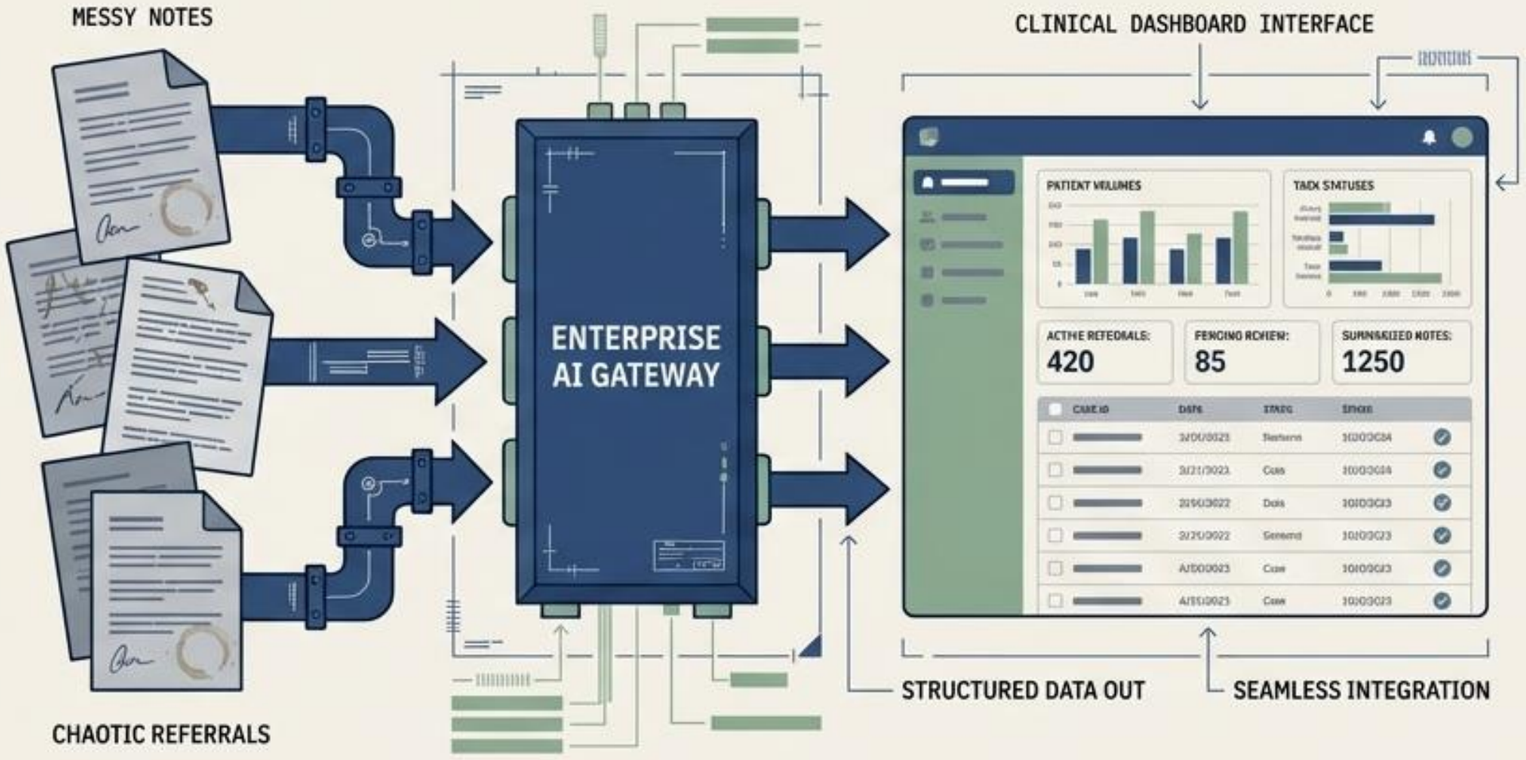


Track 1 prioritizes enterprise reliability and workflow integration

Clinicians need tools to process massive volumes of notes and referrals to make rapid patient care decisions. The focus here is on reliability, complex reasoning, and seamless workflow integration.

Example Directions:

- Referral triage assistants
- Clinical note summarization tools
- Structured information extraction
- Decision-support tools for reviewing cases
- Systems that flag missing or inconsistent information



Track 2 focuses on creative AI and novel clinical exploration

This track rewards innovative and entirely new ways of interacting with clinical information. Teams are encouraged to incorporate external public datasets to expand the capability of their solutions.

Example Directions:

- AI clinical copilots
- Case exploration tools
- Novel retrieval systems over clinical cases
- Medical knowledge assistants
- AI agents that navigate complex healthcare networks

Sparking Your Prototype Ideas

An AI assistant that summarizes patient cases.

A system extracting structured information from raw clinical notes.

A retrieval system designed to find similar patient cases.

An AI agent that analyzes lab results to flag abnormal findings.

A clinical copilot aiding in navigating comprehensive patient records.

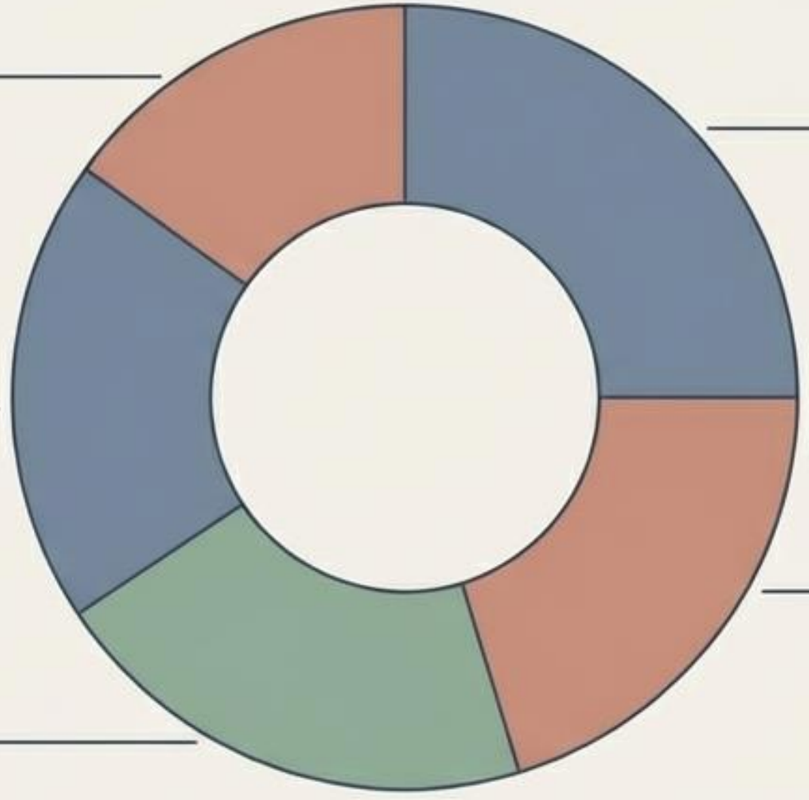
These are only starting points—explore creative ideas.

The Finish Line Evaluation Rubric

15% - Demo and Presentation:
How clearly is the technical solution demonstrated?

20% - Practical Usefulness:
Could this system realistically help in a real healthcare workflow?

20% - Reliability and Safety:
Does the system appropriately handle incomplete or ambiguous information?



25% - AI System Design:
How well is the underlying system architecture designed?

20% - Problem Understanding:
How clearly does the team define and understand the healthcare problem?

Tactical Advice for a Winning Build

Solve Clearly

Define a specific problem before you write a line of code.

Favor Function over Bloat

A simple, functional prototype always beats an overly complex, broken system.

Show Your Work

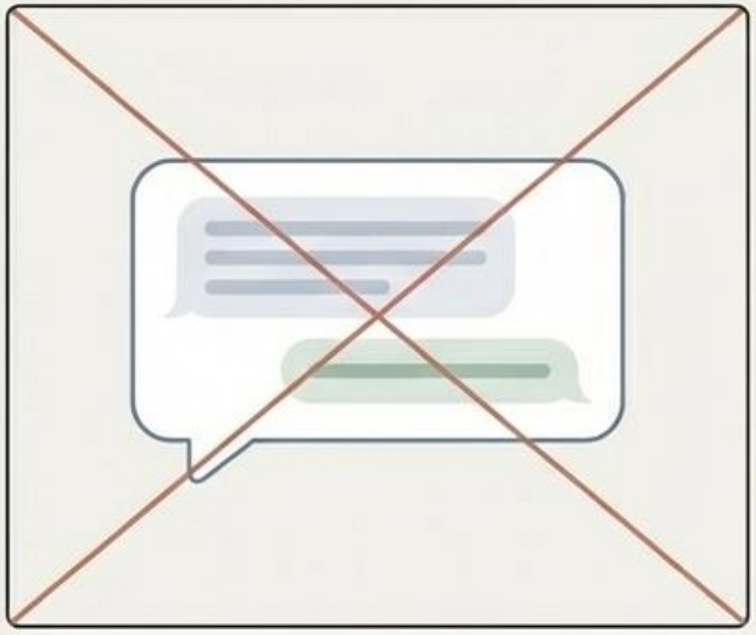
Clearly explain your architectural design decisions during the pitch.

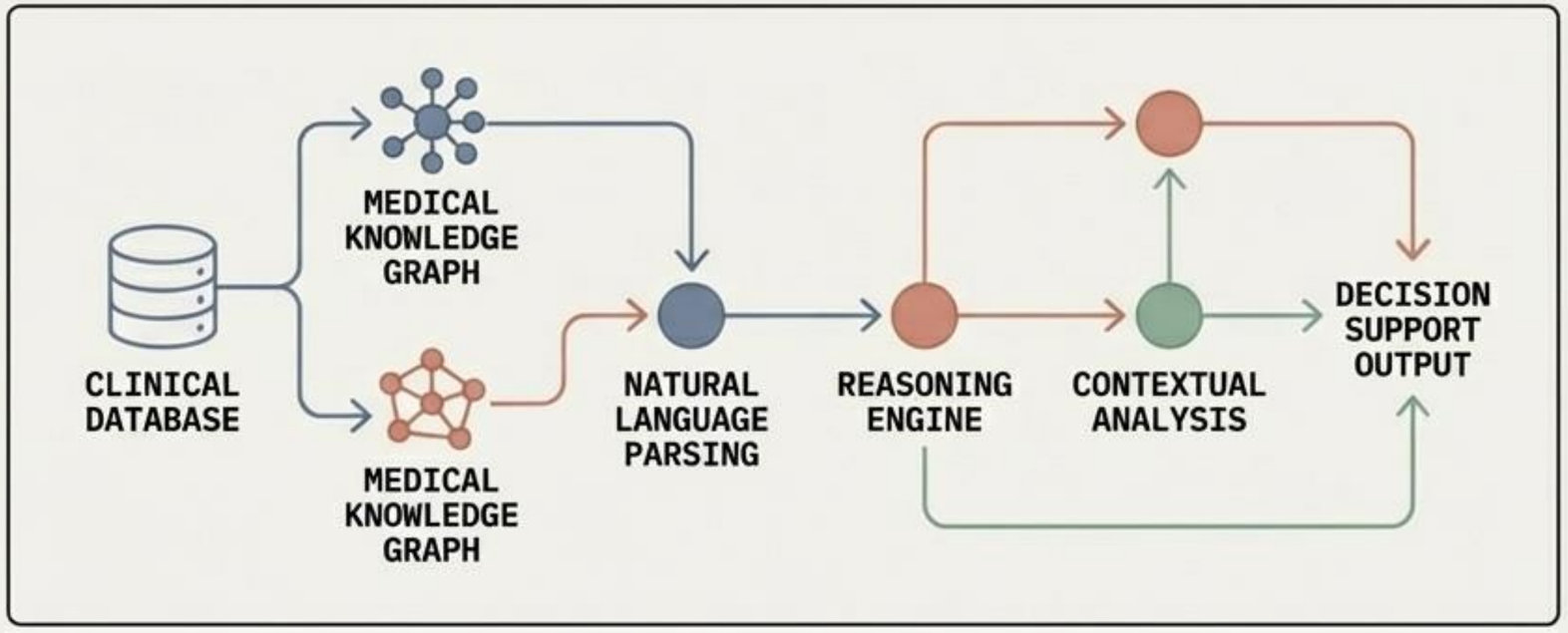
Prove It Works

Ensure your demo actually highlights how the system processes data.

Build a focused system rather than something overly complex.

Real Clinical Reasoning Requires More Than a Chatbot





EXPLORE THE DATASET.

FORM YOUR TEAMS.

START EXPERIMENTING.

"The goal is not just to build a chatbot. The goal is to explore how AI systems can genuinely understand and interpret complex healthcare information."

Logistics

- Data: https://huggingface.co/datasets/bavehackathon/2026-healthcare-ai
- Team up: max 4
- Wifi:
- In-day Demo signup: